Product Introduction

The ZDH4337 is a single-chip, fully integrated RF front-end IC. It integrates a power amplifier (PA), a low-noise amplifier (LNA), and an RF switch (SW). It typically operates from a 3.3V supply, with a maximum output power of 23dBm at the TX terminal and a noise figure of 1.8dB at the RX terminal. The input and output impedances are internally matched to 50Ω, resulting in a simple external application circuit.

The ZDH4337 is packaged in a standard green, lead-free QFN3x3-16 package, offering excellent reliability, economy, and a high cost-performance ratio.

It is suitable for improving the sensitivity and extending the transmission distance of UWB transceiver systems.

Typical application scenarios

- Precise real-time positioning systems
- Distance-aware wireless networks
- IEEE 802.15.4-2011 UWB transceivers

Absolute Maximum Ratings

Parameter	Value
Storage Temperature	-65°C~+150°C
Operating Temperature	-55°C~+125°C
Operating Voltage	6V
DC control input voltage	5V
TX RF input power	15dBm
ANT input power	10dBm

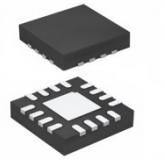
Operating status control logic table

State	TXEN	RXEN
TX on	1	0
RX on	0	1
Chip shutdown	0	0
Prohibited	1	1

"1": High level = +2.8V~+3.6V

"0": Low level = 0~+0.5V

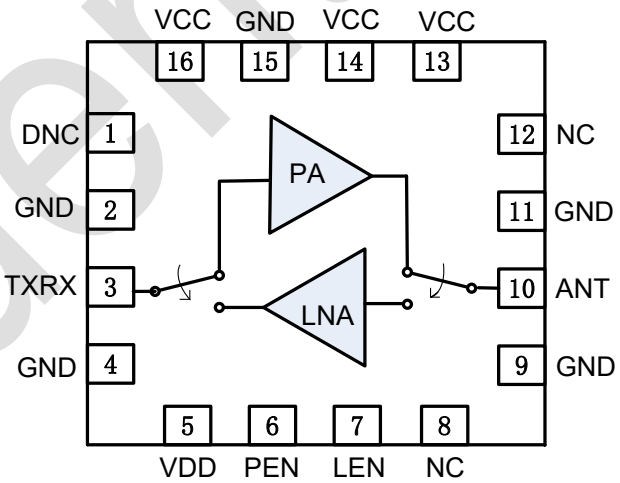
Features



- 3.3V single-voltage supply
- Small signal gain: 22dB @ 4000MHz
- Saturated output power: 23dBm @ 4000MHz, PIN=0dBm
- Single-chip IC integrating power amplifier (PA), low-noise amplifier (LNA), and two single-pole double-throw switches
- Input/output 50Ω impedance matching
- Green lead-free 16-pin QFN3x3 package

Pb This product complies with all relevant regulations and is halogen-free.

Pin diagram (Top View)



PIN No.	Pin name	Description
1	DNC	Floating
8,12	NC	Empty, floating or grounded
2,4,9,11,15	GND	Ground
3	TXRX	Transceiver RF signal input/output
5	VDD	Switching power supply
6	TXEN	Transmit enable
7	RXEN	Receive enable
10	ANT	Antenna RF signal input/output
13,14,16	VCC	PA Power supply voltage

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Recommended operating conditions

Parameter	Unit	Minimum	Typical	Maximum
Operating voltage	V	2.8	3.3	3.6
Control voltage "1"	V	2.8	-	VDD
Control voltage "0"	V	0	-	0.5
Operating temperature	°C	-40	-	85

TX 电气参数

测试条件：VDD=VCC=+3.3 V，Temp= +25°C，Freq：3000MHz~5000MHz，TXEN=3.0V，RXEN=0V，50Ω 测试系统。

参数	条件	频率					单位
频率范围(F)	-	3000	3700	4000	4300	5000	MHz
饱和输出功率(Psat)	PIN=0dBm	23	23	23	22	22	dBm
小信号增益(Gain)	-	20	22	22	21	25	dB
输入回损 (S11)	-	-11	-17	-19	-13	-13	dB
输出回损 (S22)	-	-10	-9	-4	-3	-3	dB
反向隔离（S12）	-	-31	-36	-38	-41	-40	dB
TX 静态电流 (Icq)	无 RF 输入信号	48					mA
TX 工作电流 (Icc)	PIN=0dBm	193	180	195	226	225	mA
谐波	H2,Pout=22dBm	23	28	29	24	24	dBc
	H3,Pout=22dBm	39	43	35	36	39	

RX 电气参数

测试条件：VDD=VCC=+3.3 V，Temp= +25°C，Freq：3000MHz~5000MHz，TXEN=0V，RXEN=3.0V，50Ω 测试系统。

参数	条件	频率					单位
频率范围(F)	-	3000	3700	4000	4300	5000	MHz
增益 (Gain)	-	27	23	23	23	24	dB
输入回损 (S11)	-	-1	-2	-4	-7	-11	dB
输出回损 (S22)	-	-10	-11	-10	-9	-8	dB
反向隔离（S12）	-	-36	-43	-44	-41	-35	dB
噪声系数	-	1.74	1.78	1.87	2.03	2.12	dB
RX 静态电流	无 RF 输入信号	18					mA
RX 工作电流 (I_{DD})	PIN=-10dBm	24	22	22	22	23	mA

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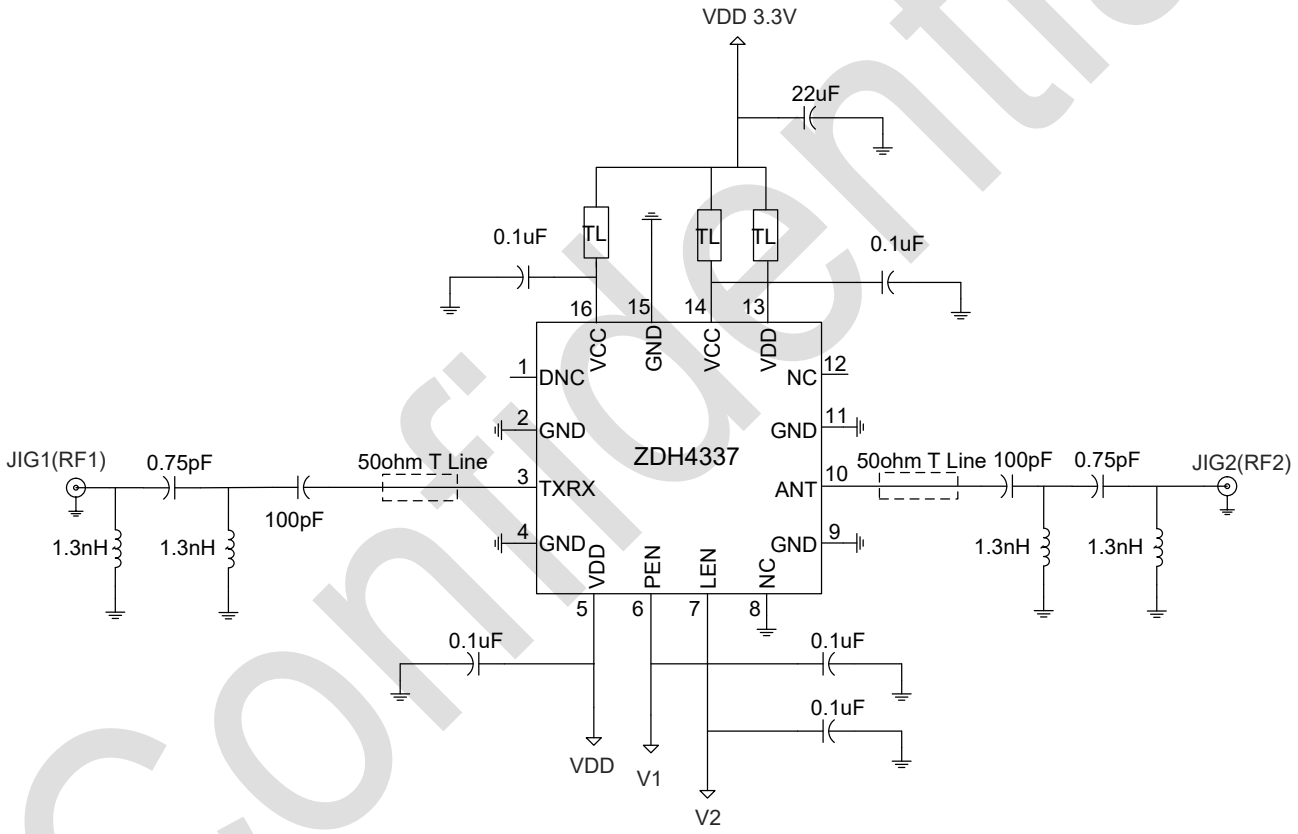
芯片关断电气参数

测试条件：VDD=VCC=+3.3 V，Temp= +25°C，TXEN=0V，RXEN=0V，50Ω 测试系统。

参数	条件	最小值	典型值	最大值	单位
关断后漏电流	-	-		1	uA
TXRX-ANT 插损	-	-	-45	-	dB
ANT-TXRX 插损	-	-	-45	-	dB
TX-RX 切换时间	10%RF -90%RF/ 90%RF-10%RF	-	300	-	ns
关断/开启切换时间	50%Vctrl-10%RF/50%Vctrl-90%RF	-	300	-	ns

应用电路图



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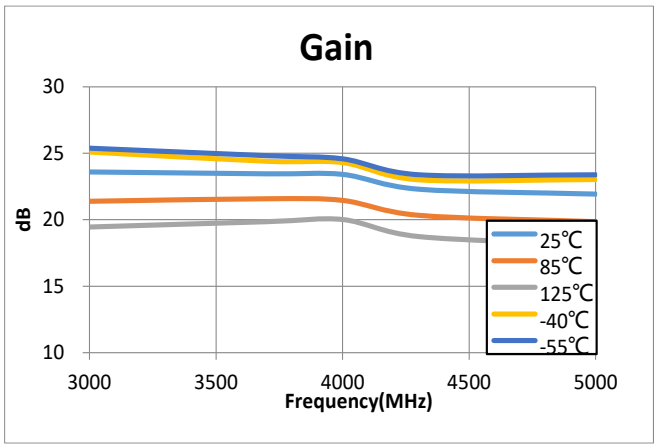
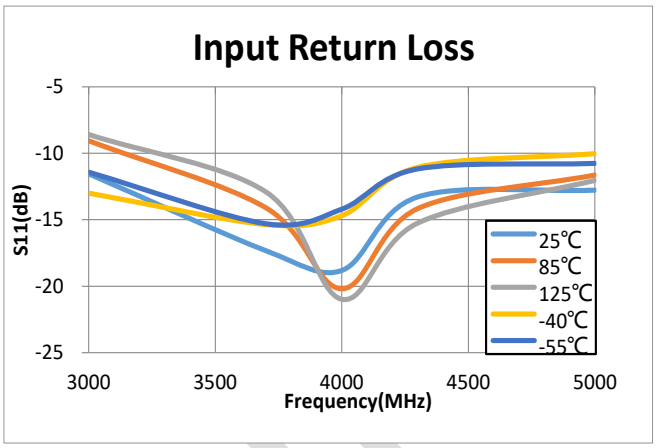
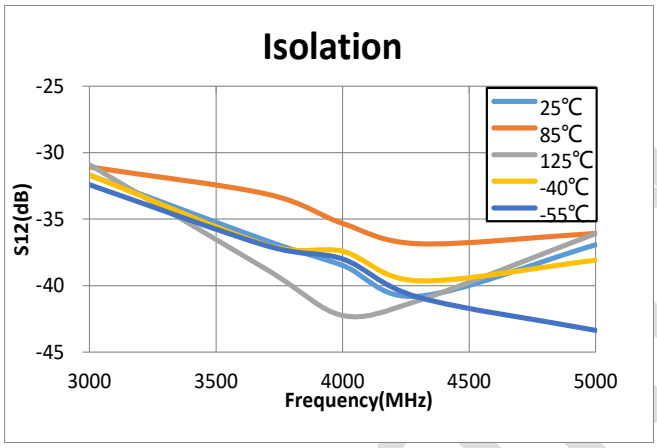
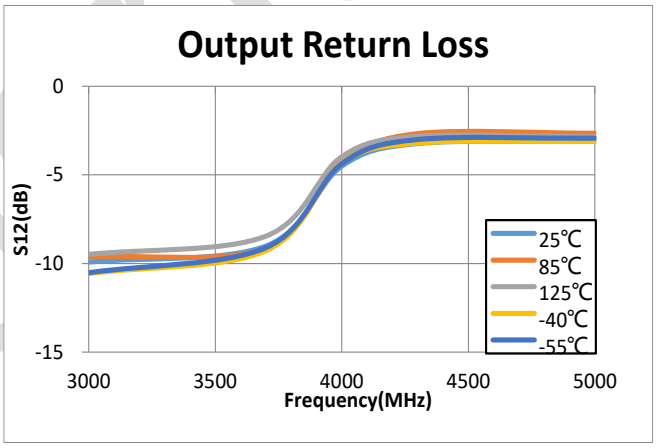
典型性能曲线图

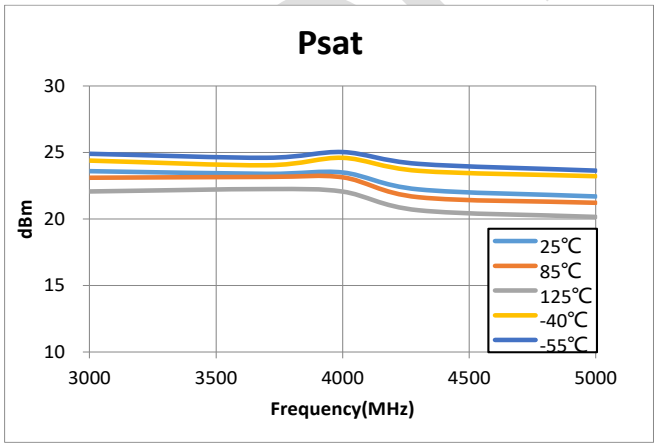
TX测试条件: VCC= VDD= +3.3V，Temp= +25°C，PIN=0dBm，TXEN=3.0V，RXEN=0V，50Ω测试系统。





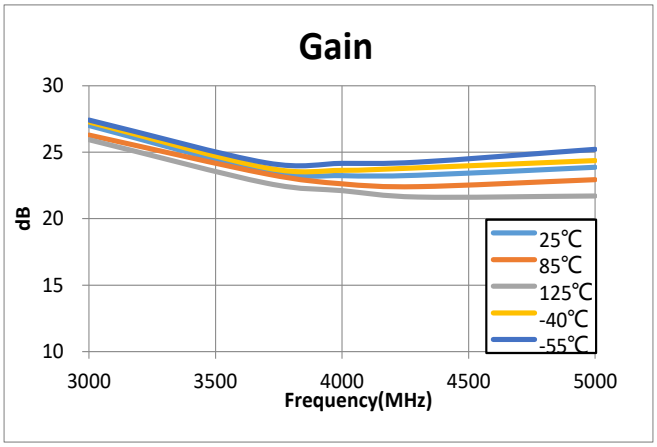


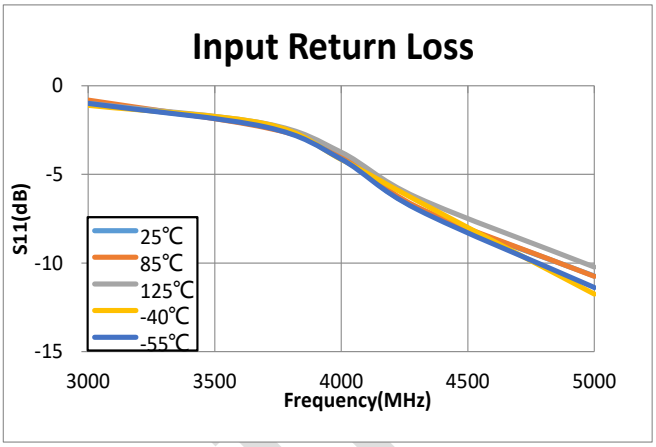


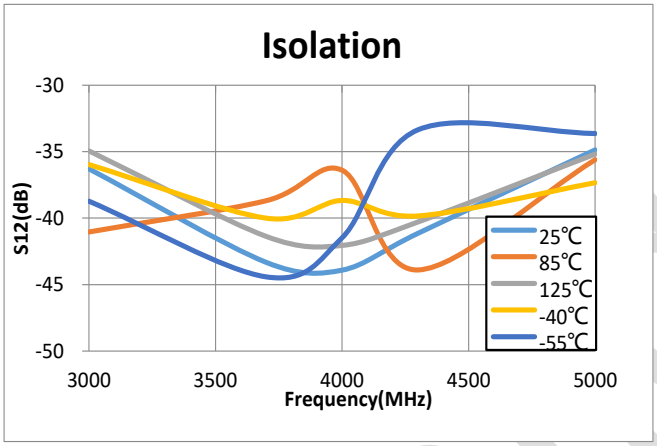


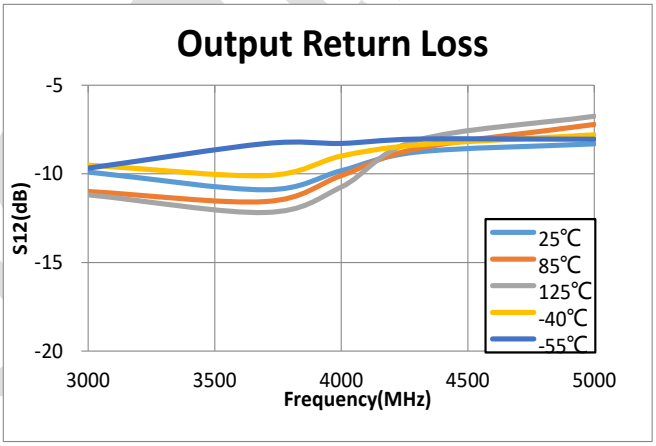
典型性能曲线图

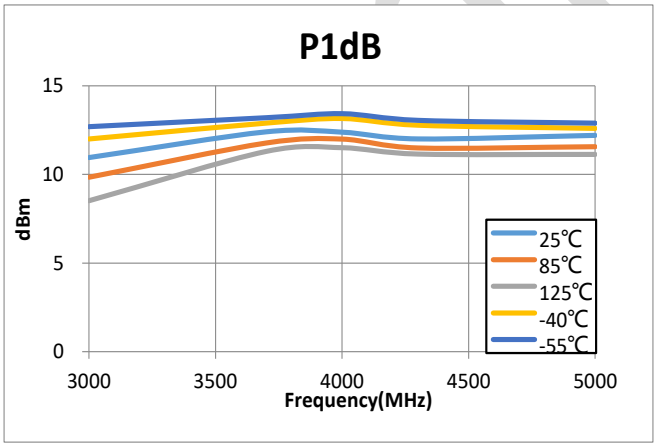
RX测试条件：VDD=VCC=+3.3 V，Temp= +25°C，PIN=0dBm，TXEN=0V，RXEN=3.0V，50Ω测试系统。

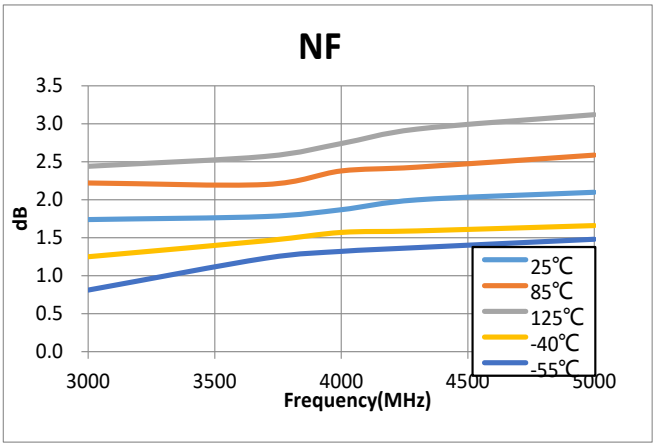




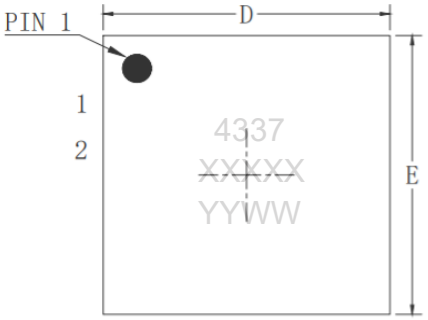




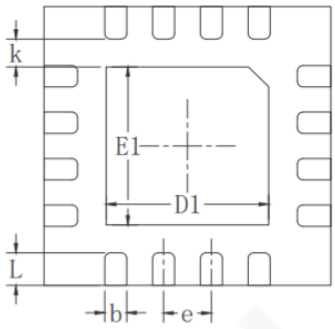




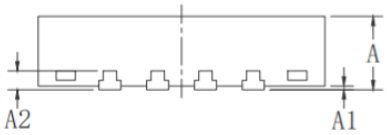
Package diagram



top view



bottom view



side view

symbol	Dimensions unit: mm		
	minimum value	Standard value	maximum value
A	0. 700	0. 750	0. 800
A1	0. 000	--	0. 050
A2	0. 195	0. 203	0. 211
D	2. 950	3. 000	3. 050
E	2. 950	3. 000	3. 050
D1	1. 600	1. 700	1. 800
E1	1. 600	1. 700	1. 800
k	0. 300 Min.		
b	0. 180	0. 230	0. 280
e	0. 500 Typ.		
L	0. 300	0. 350	0. 400

Ordering Information

Model	Marking	Packaging
ZDH4337	4337	QFN3X3-16

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